PhysInSAR: ground motion from Sentinel-1 and NISAR without phase unwrapping

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Problem

InSAR measures ground motion only as a phase that repeats every half wavelength, so every time series depends on phase unwrapping. Unwrapping fails where C-band decorrelates in vegetation and where motion is fast or spatially sharp, as over mines, landslides, subsiding cities and volcanoes, and its whole-cycle errors propagate silently into every product. With NISAR now observing the same ground as Sentinel-1 at a wavelength 4.3 times longer, a coherent but coarse L-band measurement sits next to a precise but fragile C-band one. Today the two are unwrapped separately and fused at the end, mistakes included.

Approach

PhysInSAR never unwraps. A single continuous model of three-dimensional ground motion in space and time, a neural field $\mathbf{u}_\theta(x, y, t)$, is fitted directly to the wrapped phase of both sensors (Fig. 1). A physics forward model predicts every interferogram from the motion projected onto each sensor's line of sight and wavelength, $\phi = \frac{4\pi}{\lambda}\,\boldsymbol{\ell} \cdot \Delta\mathbf{u}$, plus learned tropospheric, ionospheric (anchored by NISAR's split-spectrum estimate), orbital and DEM-error fields. A von Mises likelihood, whose concentration follows from bias-corrected coherence, compares prediction and observation; because it is periodic, a prediction that is right modulo one cycle is right. The coherent L-band phase constrains the cycle count, the precise C-band phase sharpens the estimate, and the opposite look directions of the two missions separate east from up. The relative geolocation offset between the sensors is measured from backscatter texture to about one metre, because with two geometries it cannot be learned reliably from the phase.

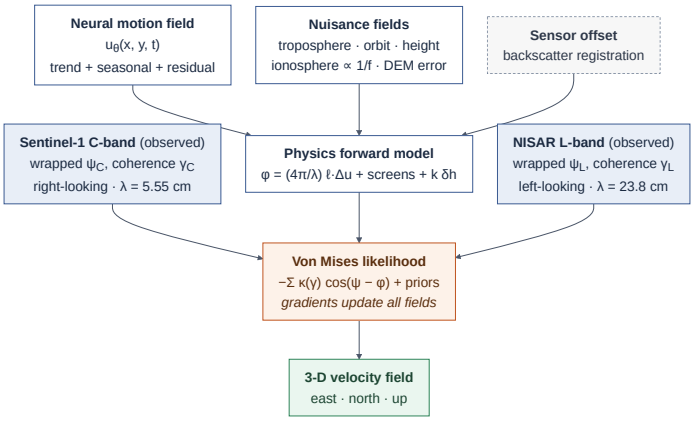


Figure 1: The observed phase of both sensors enters only through a periodic likelihood; the motion field and nuisance fields are fitted jointly.

Results on a synthetic benchmark

The method was evaluated on a paired Sentinel-1/NISAR simulator with full ground truth (single-look-complex speckle, a vegetated decorrelation belt, turbulent troposphere, L-band ionosphere, orbit and DEM errors), against the standard workflow of SNAPHU unwrapping of each sensor followed by fusion. Across a regime map of four ionosphere levels, three C-band coherence levels and five random scenes, the joint fit had the lower error in all 60 runs, including where C-band stayed coherent, with the largest gains in decorrelated vegetation (Table 1). On a sharp 700 m subsidence bowl, separate unwrapping broke down from 500 mm/yr while the joint fit stayed accurate in every scene up to 1.2 m/yr (Fig. 2). With a hidden 100 m offset between the sensors, backscatter registration restored the joint fit's accuracy (9.5/7.4 against 14.8/12.0 mm/yr without it).

Table 1: East/up velocity error (mm/yr). Regime map: mean of 20 runs per row; subsidence: median of five scenes.

Experiment	Condition	Joint fit	Separate	Joint better
Regime map, vegetated belt	C-band coherence 0.05	**9.8 / 10.1**	16.1 / 12.2	
	C-band coherence 0.30	**8.5 / 6.1**	11.0 / 8.0	
	all conditions	**9.0 / 7.7**	13.1 / 9.7	60 / 60 runs
Sharp subsidence bowl	800 mm/yr	**18.2 / 9.9**	77.4 / 61.0	5 / 5 scenes
	1200 mm/yr	**25.4 / 11.9**	169.6 / 130.6	5 / 5 scenes
Broad subsidence bowl	1200 mm/yr	33.1 / 29.2	**27.8 / 13.4**	1 / 5 scenes

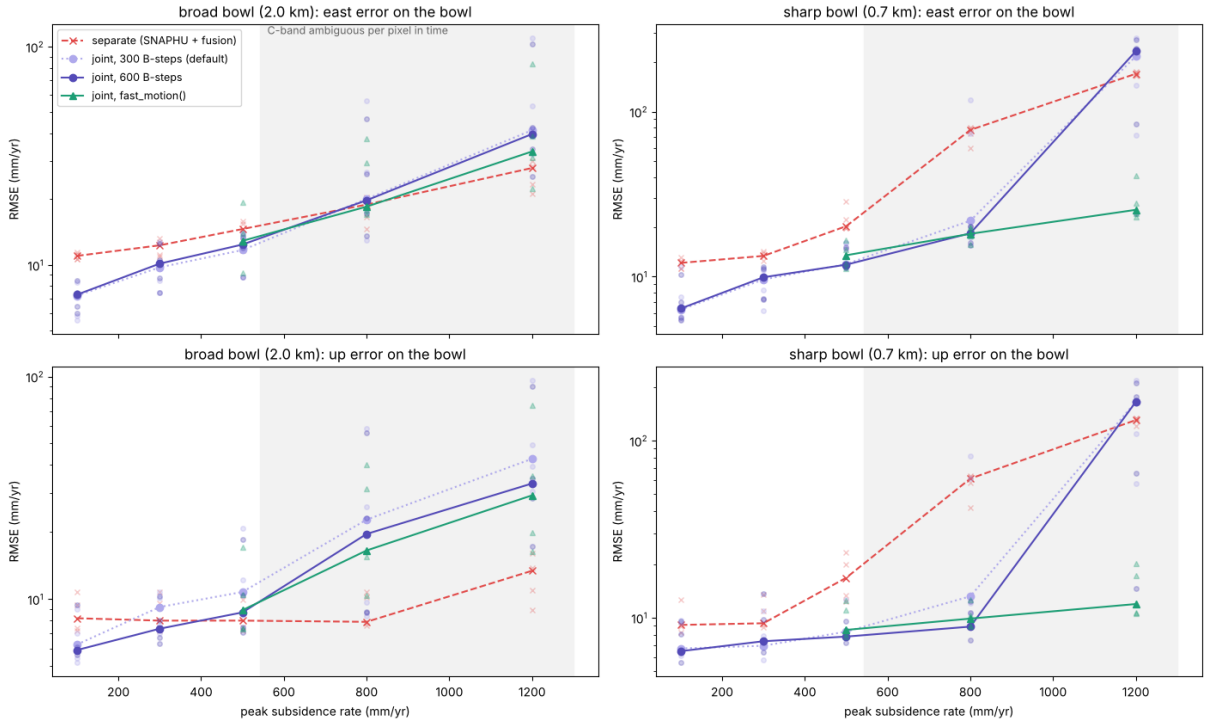


Figure 2: East (top) and up (bottom) error on the bowl versus subsidence rate for a broad 2 km (left) and a sharp 700 m (right) bowl. Red: separate unwrapping; purple: joint fit with default settings; green: joint fit with `fast_motion` settings; shading: beyond C-band's per-revisit limit; dots: individual scenes.

Limitations and open questions

All results so far are synthetic, from a simulator that shares the physics of the model, so the margins should be read as an upper bound until confirmed on real data. On broad, very fast motion, spatial unwrapping keeps up and gives the better vertical component above about 800 mm/yr, and strongly stratified troposphere over high relief is not yet handled by default. Ablations with C-band only and L-band only are still needed to separate the benefit of combining the wavelengths from that of avoiding unwrapping. The main physical risks for real data are residual L-band ionosphere and phase effects of vegetation and soil moisture, which act differently at the two wavelengths.

Status and next steps

The real-data pipeline is in place: NISAR L2 GUNW products over Mexico City (ascending path 19, frame 11, nine pairs from June to October 2026) have been downloaded through the open-source package pygeofetch, and Sentinel-1 interferograms are formed with its InSAR chain without phase filtering. The next steps are the Mexico City inversion with comparison against GNSS and published rates, the ablations, a vegetated tropical test site (for example the mining belt of Ghana), and a first presentation in early 2027, followed by a journal paper in IEEE TGRS.

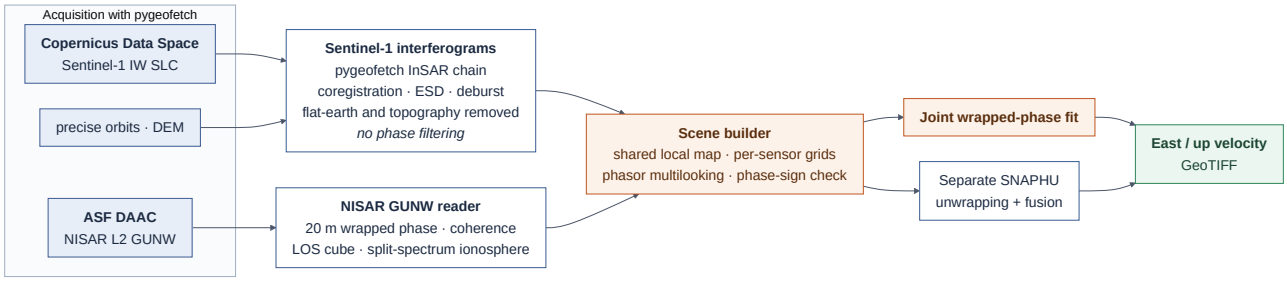


Figure 3: Real-data chain for the Mexico City case study: acquisition with pygeofetch, unfiltered Sentinel-1 interferograms and NISAR GUNW products on per-sensor grids, joint inversion, and separate SNAPHU unwrapping as reference.